

Sharif University of Technology
Department of Mechanical Engineering

Velocity Kinematics of Robots

Robotics Course – Homework 4

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1 Problem Statements

The coding for the requested problem statements is carried out in the MATLAB environment.

1.1 Question 1

The requirement of the first question is a function in MATLAB that, as input, receives the Denavit–Hartenberg table and the joint-variable vector Q symbolically, and returns the translational Jacobian matrix (J_v) and the rotational Jacobian matrix (J_ω) symbolically as output.

It is clear that we must first use the Denavit–Hartenberg table to obtain the homogeneous transformation matrix from frame n to frame 0 (H_n^0), and then proceed to compute the Jacobian matrices.

Then, at the end of the problem, we are asked to compute the Jacobian matrices of the robots given in Homework Sets 2 and 3 using the function that was derived.

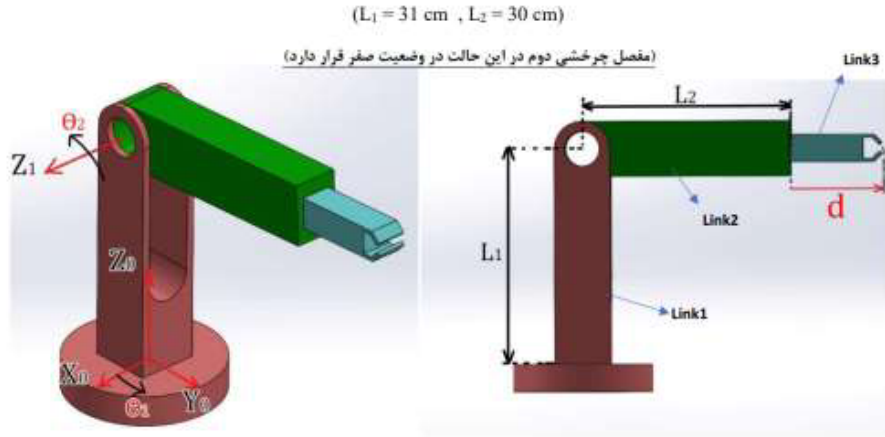
1.2 Question 2

This is a bonus question. In this question we are asked to find any singular points that exist in the robots of Homeworks 2 and 3, if such points exist, and to obtain them parametrically.

1.3 Question 3

In this question we consider the spherical robot of Homework 3 and assume that its end-effector must be able to produce the velocity vector $\vec{v} = (150, 100, 50)$ mm/s at every point. The robot's workspace is taken such that the first joint (θ_1) moves from 0° to 80° , the second joint (θ_2) from -45° to 45° , and the third joint moves from $d = 100$ mm to $d = 200$ mm.

The maximum required velocity for each of the joints is what this problem asks for.



(In this configuration, the second revolute joint is at its zero position; $L_1 = 31$ cm, $L_2 = 30$ cm.)

2 Answers to the Questions

2.1 Answer to Question 1

To answer this question, we create a function in MATLAB that receives the Denavit–Hartenberg table (DH) and the joint-variable vector (Q) as inputs, and whose output is $[J_v, J_\omega]$.

First, using the given Denavit–Hartenberg table and the functions written in Homework 1 (`Rot` and `Trans`), we obtain the homogeneous transformation matrices of the coordinate frames. As we know, in general, the homogeneous transformation matrix takes the following form:

$$H_n^0 = \begin{bmatrix} R_n^0 & \overrightarrow{O_0O_n} \\ 0 & 1 \end{bmatrix}$$

As can be seen, the rotation matrix and the position vector of the end-effector can be extracted from the homogeneous transformation matrix.

Then, using the relations given in Lecture 11, we obtain the translational and rotational Jacobians by the analytical method.

The procedure for computing these two matrices is described in the function below:

Listing 1: `JcbCalc.m`

```
function [Jv, Jw] = JcbCalc(DH, Q)

s = size(DH);
n = s(1);
H = sym(eye(4));

for k = 1:n
    H = H*Rot('Z',DH(k,1))*Trans('Z',DH(k,2))* Trans('X',DH(k,3))*Rot('X', DH(k,4));
end

R = H(1:3, 1:3);
O0On = H(1:3, 4);

Jv = sym(eye(3,n));

for k = 1:3
    for t = 1:n
        Jv(k,t) = diff(O0On(k), Q(t));
    end
end

Jw = sym(eye(3,n));
for k = 1:3
    for m = 1:n
        Jw(k,m) = 0;
        for j = 1:3
            switch k
                case 1
                    Jw(k,m) = Jw(k,m) + diff(R(3,j), Q(m))*R(2,j);
                case 2
                    Jw(k,m) = Jw(k,m) + diff(R(1,j), Q(m))*R(3,j);
                case 3
                    Jw(k,m) = Jw(k,m) + diff(R(2,j), Q(m))*R(1,j);
            end
        end
    end
end
end
```

end

Now, to compute the Jacobian matrices for Homeworks 2 and 3, we need the Denavit–Hartenberg tables of the two robots given in these assignments.

From the Homework 2 report, the Denavit–Hartenberg table of the robot is as follows:

	θ	d	l	α
1	θ_1	d_1	0	0
2	-90	L_2	$-L_1$	-90
3	0	$d_2 + L_3$	0	0

In the table above we have:

$$L_1 = 45 \text{ mm}, \quad L_2 = 22.5 \text{ mm}, \quad L_3 = 30 \text{ mm}$$

$$Q = \begin{bmatrix} \theta_1 \\ d_1 \\ d_2 \end{bmatrix}$$

Similarly to the above, the Denavit–Hartenberg table and the joint-variable vector of the spherical robot (Homework 3) – also taken from the Homework 3 report – are as follows:

	θ	d	l	α
1	θ_1	310	0	0
2	90	0	0	90
3	θ_2	0	$300 + d$	0

$$Q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d \end{bmatrix}$$

We implement the following MATLAB code to compute the Jacobian matrices of the two robots above:

Listing 2: HW4.m

```
clc; clear;

%Robot 2
syms theta1;
syms d1;
syms d2;

DH2 = [theta1, d1, 0, 0;
       -90, 22.5, -45, -90;
       0, d2 + 30, 0, 0];

Q2 = [theta1; d1; d2];

[Jv_2, Jw_2] = JcbCalc(DH2,Q2);

Jv_2
Jw_2
```

```

%Robot 3
syms theta1;
syms theta2;
syms d;

DH3 = [theta1, 310, 0, 0;
       90, 0, 0, 90;
       theta2, 0, 300+d, 0];

Q3 = [theta1; theta2; d];

[Jv_3, Jw_3] = JcbCalc(DH3, Q3);

Jv_3
Jw_3

```

The result is as follows:

```

Jv_2 =
[- (pi*cos((pi*theta1)/180))/4 - (pi*sin((pi*theta1)/180)*(d2 + 30))/180, 0,
cos((pi*theta1)/180)]
[ (pi*cos((pi*theta1)/180)*(d2 + 30))/180 - (pi*sin((pi*theta1)/180))/4, 0,
sin((pi*theta1)/180)]
[ 0, 1,
0]

Jw_2 =
[ 0, 0, 0]
[ 0, 0, 0]
[(pi*cos((pi*theta1)/180)^2)/180 + (pi*sin((pi*theta1)/180)^2)/180, 0, 0]

Jv_3 =
[-(pi*cos((pi*theta1)/180)*cos((pi*theta2)/180)*(d + 300))/180,
(pi*sin((pi*theta1)/180)*sin((pi*theta2)/180)*(d + 300))/180, -
cos((pi*theta2)/180)*sin((pi*theta1)/180)]
[-(pi*cos((pi*theta2)/180)*sin((pi*theta1)/180)*(d + 300))/180, -
(pi*cos((pi*theta1)/180)*sin((pi*theta2)/180)*(d + 300))/180,
cos((pi*theta1)/180)*cos((pi*theta2)/180)]
[ 0,
(pi*cos((pi*theta2)/180)*(d + 300))/180, sin((pi*theta2)/180)]

Jw_3 =
[
0, (pi*cos((pi*theta1)/180)*cos((pi*theta2)/180)^2)/180 +
(pi*cos((pi*theta1)/180)*sin((pi*theta2)/180)^2)/180, 0]
[
0, (pi*cos((pi*theta2)/180)^2*sin((pi*theta1)/180))/180 +
(pi*sin((pi*theta1)/180)*sin((pi*theta2)/180)^2)/180, 0]
[(pi*cos((pi*theta1)/180)^2)/180 + (pi*cos((pi*theta2)/180)^2*sin((pi*theta1)/180)
^2)/180 +
(pi*sin((pi*theta1)/180)^2*sin((pi*theta2)/180)^2)/180,
0, 0]

```

** The related codes and files are included in the appendix.*